River Deep Mountain AI Use Case

Use case 2: How might we support integrated catchment planning, enabling effective evidence sharing across stakeholders?

River Deep Mountain AI

Version overview

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Approval

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Executive Summary

This report outlines a use case demonstrating how the River Deep Mountain AI (RDMAI) model suite, specifically the Open Risk Map, Open Flow and Open Orthophosphate models could be applied to support integrated catchment planning. By providing a unified, open-source evidence base, these tools help to enable a range of stakeholders to move beyond siloed decision-making and co-design cost-effective strategies for improving water body health.

The recent Corry Review and Cunliffe Report emphasise the urgent need for integrated planning, where decision-making is underpinned by shared, interoperable data rather than isolated sectoral targets. However, effective collaboration is often hindered by data deficits, technical fragmentation, and a lack of trust in proprietary black box models.

This report details a hypothetical workflow where a Catchment Partnership uses RDMAI models to address phosphorus failure in a river catchment. The suggested process integrates spatial, hydrological, and chemical data as a tool to develop a catchment management plan.

The report concludes by breaking down the value proposition for the model-led workflow, discussing the potential strategic benefits of deploying the RDMAI models as decision support tools.

This report details the second use case for the River Deep Mountain AI (RDMAI) project, exploring how open-source artificial intelligence and machine learning models can drive more collaborative and evidence-based water management. By combining the RDMAI Open Risk Map, Open Flow, and Open Orthophosphate models, the report demonstrates a workflow for unifying datasets into a shared evidence base. This approach aims to bridge the gap between technical modelling and strategic decision-making, enabling water companies, regulators, and partners to co-design effective interventions for catchment health.

1 Supporting integrated catchment planning

Water companies, catchment managers, and regulatory bodies face significant organisational and financial complexities when attempting to design and implement integrated strategies for a river catchment. Even within a single organisation, it can be difficult to reconcile competing priorities, such as connecting new housing developments while reducing nutrient loading. As such, there is a requirement for a unified, evidence-based view of the system.

Integrated Catchment Planning is a means to bridge these operational silos. There are a range of motivations for this shift, from the statutory requirements set out in regulation to the need to ensure that every pound spent delivers the maximum environmental benefit and value to customers.

1.1 Regulatory relevance of integrated catchment management

There are several regulatory requirements and statutory frameworks in the UK with links to integrated catchment management. These requirements primarily fall on water companies, environmental regulators and local authorities.

1.1.1 Water Resources Management Plans

Under the Water Industry Act 1991, water companies in England and Wales have a statutory duty to prepare and maintain Water Resources Management Plans (WRMPs) on a five-yearly basis.

The plans set out how the water company will achieve a secure water supply while protecting and enhancing the environment. The plans may require nature-based solutions or upstream catchment management to address pollution at the source, rather than relying solely on treatment processes prior to distribution.

1.1.2 Water Framework Directive

Retained via the *Water Environment (Water Framework Directive) (England and Wales) Regulations 2017*) (UK Government, 2017), the Water Framework Directive (WFD) requires the production of River Basin Management Plans (RBMP). RBMPs are produced every six years by the relevant appropriate authority (e.g. the Environment

Agency (EA) in England) and set the environmental objectives for all water bodies within each river basin district.

The primary objective is to achieve 'Good Status' (European Commision, n.d.) for each water body within a river basin district, a metric that combines biological health, chemical quality, and hydromorphology. To achieve this, RBMPs must outline a statutory 'Programme of Measures', a schedule of actions required to meet these targets (Environment Agency, 2024). The regulations mandate an integrated approach to managing water quality and quantity, ensuring that neighbouring authorities cooperate to manage shared river basins. Furthermore, the Directive enforces a 'no deterioration' principle, requiring monitoring to ensure that current water body status is maintained alongside any planned improvements.

1.1.3 Drainage and Wastewater Management plans

Under the *Environment Act 2021* (UK Government, 2021), all water and sewerage companies in England and Wales are required to produce Drainage and Wastewater Management Plans (DWMPs). DWMPs cover a minimum 25-year period, setting out how wastewater systems will be maintained, improved, and extended to cope with the challenges of climate change, population growth and urban creep. DWMPs aim to ensure that wastewater risks are not managed in isolation but are understood in the context of wider catchment pressures.

DWMPs use a hierarchical planning structure, consisting of three levels, each at a different spatial focus. Level 1 provides a strategic, water company-wide baseline of risk. Level 2 involves planning at the wastewater catchment scale, aggregating risks from multiple distinct sewer networks to identify systemic vulnerabilities. The approach promotes the selection of catchment-based solutions, such as sustainable drainage systems or natural flood management. Finally, Level 3 develops site-specific interventions.

1.1.4 Flood Risk Management Plans

Flood Risk Management Plans (FRMPs) serve as strategic frameworks designed to mitigate the adverse consequences of flooding on human health, the environment, cultural heritage, and economic activity (Environment Agency, 2022). Their primary purpose is to outline the specific objectives and measures required to manage flood risks within a designated river basin or coastal district over a set cycle (typically six years). By integrating data from flood hazard and risk maps, these plans help authorities identify high-risk areas and prioritise investments in flood defences, natural water retention measures, and early warning systems.

Beyond immediate infrastructure, FRMPs facilitate a coordinated approach to water management involving various stakeholders, including local authorities, environmental agencies, and the public. They aim to ensure that flood management strategies are not developed in isolation but are aligned with RBMPs and climate

change adaptation strategies. This holistic approach helps communities to build long-term resilience.

1.2 The Catchment Based Approach (CaBA)

The Catchment Based Approach (CaBA) is a policy framework launched by Defra in 2013 that sets out a method for delivery of integrated catchment planning at the local level, often producing catchment management plans (UK Government, 2013). While regulations like the WFD set the targets and legal obligations, the CaBA provides the collaborative governance structure required to deliver them. CaBA aims to create a forum where the operational silos between organisations are dismantled, allowing NGOs, water companies, local authorities and landowners to collaborate more effectively. CaBA partnerships help facilitate the work required to resolve water body health challenges, satisfying the priorities of multiple organisations simultaneously in a coordinated way.

1.3 The future direction of integrated catchment planning

Recent high-level reviews have moved beyond merely suggesting better cooperation, instead calling for a fundamental restructuring of how the water sector is governed to enforce integration. The findings of the Independent Water Commission (2025) provide the strongest mandate yet for this systemic overhaul, arguing that the current regulatory architecture is no longer fit for purpose.

1.3.1 Cunliffe report

The Independent Water Commission's final report sets out recommendations for reform to improve the water sector regulatory system in England and Wales. The Cunliffe Report (Independent Water Commission, 2025) serves as a mandate for integrated catchment management, explicitly recommending the replacement of the current fragmented planning regime with a 'Comprehensive Systems Planning Framework' (recommendation 3). To deliver this, the Commission proposes the creation of new Regional Water Authorities in England (and a single national authority in Wales) to oversee long-term investment. These bodies would be empowered to direct funding across sectoral boundaries, breaking the 'silo' culture where water companies, farmers, and NGOs operate independently. Crucially, the report argues that Catchment Partnerships must be formally reinforced and integrated into this regional decision-making structure, moving them from voluntary bodies to central pillars of the delivery network. This aligns with the Commission's call for a Single Integrated Regulator (recommendation 16) to unify the functions of Ofwat, the EA, and the Drinking Water Inspectorate (DWI), ensuring that environmental outcomes and economic regulation are joined up.

1.3.2 Corry review

Dan Corry's review examines whether Defra's regulatory landscape (the regulators and regulations) is fit for purpose in driving both economic growth and nature

recovery. The Corry Review (Defra, 2025) makes a strong case for shifting from a rigid, siloed regulatory model to a 'systems-level' approach that prioritises outcomes over process. A key recommendation in favour of improved catchment planning is the empowerment of Catchment Partnerships and the establishment of Regional Systems Planners. The review suggests that these local partnerships should be better recognised and potentially better funded to develop high-quality 'local catchment management plans'. These plans would not sit in isolation but would form the foundational 'building blocks' for national strategies, ensuring that top-down targets (such as those in the Environmental Improvement Plan (Defra, 2025)) are delivered through locally tailored, outcome-focused interventions. Furthermore, the report recommends granting 'trusted partners' (such as Rivers Trusts or farmer clusters) greater autonomy to deliver nature recovery projects, reducing the 'green tape' that currently hinders collaborative, catchment-scale action.

However, the review explicitly identifies that this integrated future is currently obstructed by significant 'data deficits' and technological fragmentation. It highlights that the current regulatory system is held back by 'outdated IT systems' and 'data limitations', where critical information is often siloed within different agencies or is simply not up to date. The report argues that for a risk-based, outcome-focused approach to work, there must be a move towards collaborative monitoring and a more unified approach. Without modernising these digital infrastructures to make monitoring data accessible and interoperable between water companies, regulators, and catchment partners, the ability to make evidence-based decisions at a catchment scale remains severely compromised.

2 The challenge of integrated catchment planning

Water companies, catchment managers, and regulatory bodies face significant organisational, technical and financial complexities when attempting to design and implement integrated strategies for a river catchment.

As evidenced in the Corry review, a fundamental barrier to integrated catchment planning is the deficit of observed data, particularly in upstream environments. While the regulatory framework demands an evidence-based approach, the physical reality is that stakeholders are often attempting to manage complex systems with incomplete information.

Further, and as noted in the Cunliffe Report, regulatory sampling is often limited to specific points and low frequencies (e.g., monthly spot samples). This routine monitoring regime frequently misses short-term, high-impact events such as storm runoff or Storm Overflow (SO) spills. For an integrated planner, this creates a blind spot. There is a disconnect between the continuous nature of river dynamics and the intermittent nature of data collection.

To navigate these challenges, stakeholders require effective tools to support integrated catchment planning.

3 Using the RDMAI models to support integrated catchment planning

The following use case demonstrates how three of the eight RDMAI models (Open Risk Map, Open Flow and Open Orthophosphate) can be combined to help stakeholders to navigate the complexities of integrated catchment management. By delineating risk beyond the spatial limitations of existing monitoring networks, the RDMAI models provide a level of catchment visibility essential for understanding water body health.

The RDMAI suite of models offer the advantage of being fully open source. In a multi-stakeholder environment, such as a CaBA partnership, transparency builds trust. By allowing regulators, water companies, and third parties to inspect and validate the underlying logic, the models function as a 'glass box' rather than a proprietary 'black box.' This shared visibility ensures that all partners are working from a common, agreed-upon evidence base.

Further details on each of the models discussed can be found in their respective model documentation. Section 8 outlines key model limitations that users should be aware of.

3.1 The Open Risk Map model

The Open Risk Map is built up of individual risk layers, each of which aligns with a particular risk type (orthophosphate-P and ammoniacal-N) and source pressure informed by the EAs reasons for not achieving good (RNAG) status, including:

- Sewage discharge (continuous), i.e., Wastewater Treatment Works (WwTWs) final effluent discharges
- Sewage discharge (intermittent), i.e., overflow spills
- Agricultural runoff
- Urban runoff
- Private sewage treatment.

Each risk layer uses openly available data to calculate a concentration, which is added to a baseline catchment map. The Open Risk Map then determines how the level of risk varies throughout the river network in a catchment. Risk Map outputs compare reasonably well against current industry-standard modelling software (SAGIS) – see the benchmarking report.

3.2 The Open Flow model

The RDMAI Open Flow model is a machine learning (ML) model designed to estimate daily mean river flow (m^3/sec) in ungauged catchments. Built on a Long-Short-Term Memory (LSTM) architecture within a recurrent neural network (RNN), the model integrates both static (catchment attributes which rarely change) and dynamic data sources, including historical flow, weather, and catchment attributes from the CAMELS-GB dataset, the EA's Hydrology Data Explorer, and a curated list of chalk streams from Natural England.

The CAMELS-GB dataset spans 40 years of daily river flow records, including 409 catchments across England, enabling the model to learn complex relationships between weather patterns, topography, and catchment characteristics. Time-series data are processed through an LSTM layer using a 30-day window to capture antecedent effects on flow, while static attributes are handled separately. The outputs from these two branches are then combined to produce a time-series flow prediction for each catchment.

3.3 The Open Orthophosphate model

The RDMAI Open Orthophosphate model is based on an Artificial Intelligence (AI)/ ML framework using historical observations from the EA water quality archive. The model helps define the relationships between the water quality determinands in the EA database and the orthophosphate concentration in rivers, providing orthophosphate concentration estimates, including for sites where orthophosphate is not currently monitored.

The model is built using LightGBM (Light Gradient Boosting Machine), an algorithm based on gradient boosting decision trees that captures non-linear relationships in data. The model uses SHAP (Shapley Additive exPlanations) to explain the model predictions by quantifying the contribution of each input feature to the model results.

3.4 Data sources required

To run the three models, various data sources are required (Table 1). Nearly all data used in the models are open source and free to download. As the RDMAI models are open source, the user could amend the models to enable the use of alternative data. This use case focuses on the data sources used in the development of each model.

Table 1 Data inputs required to run the RDMAI models

RDMAI Model	Data Sources
Open Risk Map	The EA consents database needs to be manually downloaded into a stipulated folder. All other datasets are processed automatically by the provided script. • Rivers: Open River Network • Overflow spill data: EA Storm Overflows - Annual Returns • Consents data: EA Consents Database • Land use: CROME 2023 dataset • Hydrology of Soil Types (HOST) derived from this study. • Rainfall data: UK Met Office Rainfall Data • Roads and buildings: OS OpenMap - Local • Built-up areas: OS Open Built Up Areas • Farmscoper Data: ADAS Farmscoper Tool is used for agricultural and livestock runoff estimates via lookup tables.
Open Flow Model	• Catchment attributes: CAMELS-GB • EA hydrology data: Hydrology Data API • Natural England chalk stream locations: Chalk streams in England:
Open Orthophosphate Model	Prior to running the model, the data needs to be pre-processed. • Up to 45 determinands available in the EA water quality archive in timeseries form, (the model can still function with fewer than 45 determinands) • Location of the sample, day of the week and month of the year, and season.

4 Workflow

4.1 Model set up and stakeholder requirements

Users will need to follow the instructions provided in the model GitHub repositories to download and run the models.

4.1.1 Open Risk Map Model

The RDMAI Open Risk Map is designed to be highly accessible, with both the model and its input data freely available. Its setup is straightforward and requires basic

Python skills, such as cloning a GitHub repository, configuring a Conda environment, and working with Jupyter Notebooks. The model is largely 'plug-and-play', enabling stakeholders to run analyses with minimal technical experience. Once the environment is prepared, running a catchment typically takes between 5 and 30 minutes on a standard laptop, depending on the catchment size. This makes the Risk Map model well-suited for users with limited coding experience who need quick and reliable outputs, that are repeatable for multiple catchments.

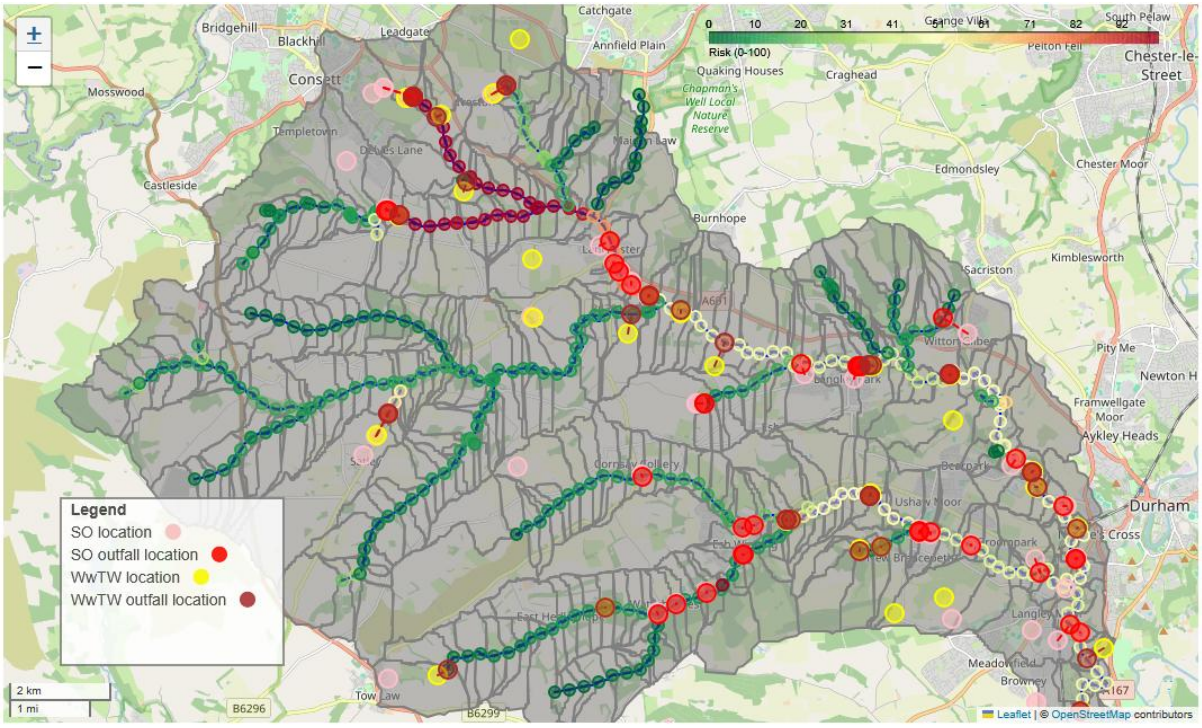

Figure 1 Risk distribution for the River Browney catchment, County Durham

4.1.2 Open Flow Model

The RDMAI Open Flow model involves a comparably more complex setup process. Installing the model and its dependencies, downloading and preparing the necessary datasets, and running the trained model for a prepared example generally takes 1 to 2 hours. Most of this time is spent on preparing the catchment datasets, while executing the trained model itself takes only a few seconds once data preparation is complete. The stakeholder could train the model to their specific catchment, but this would involve additional time and expertise. Due to these requirements, the Flow Model is best suited for stakeholders with intermediate Python skills and familiarity with scripting and data handling. While the initial setup is time-intensive, subsequent runs are fast, making it efficient for repeated use.

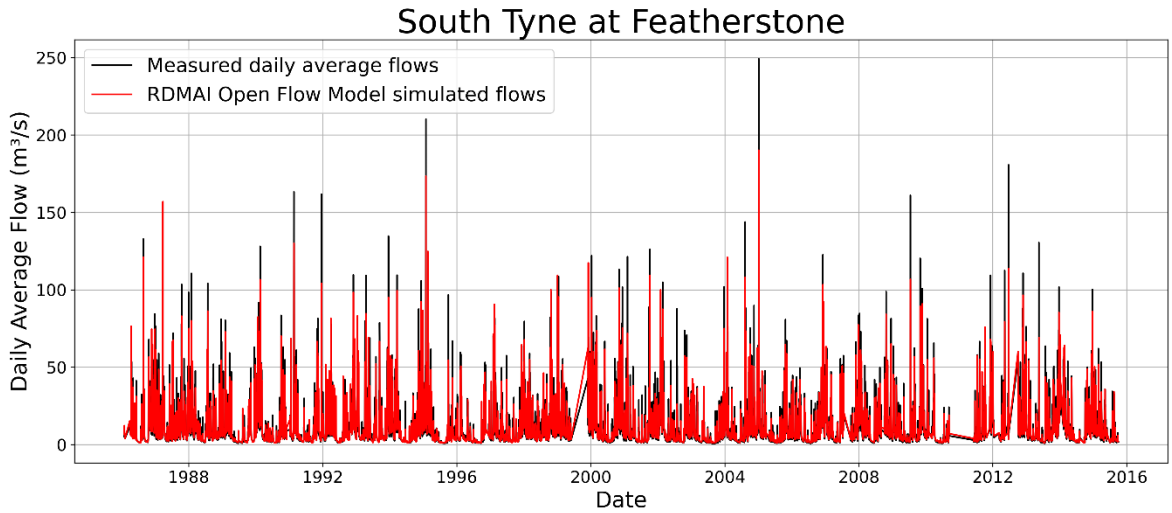

Figure 2 Timeseries plot showing validation output of the RDMAI Open Flow Model versus gauged flows. This example, for the gauging site 'South Tyne at Featherstone' in North East England, represents an instance in which the RDMAI flow model performed relatively well, achieving a Nash-Sutcliffe Efficiency (NSE) of 0.89

4.1.3 Open Orthophosphate Model

The RDMAI Open Orthophosphate model is open source and free to use. The model was developed using data from the EA water quality archive. Considerable time (days) is required to process the input data to create additional features and format the data as required for the model. An experienced Python programmer is required to understand and perform the steps for input data preparation and to run the model.

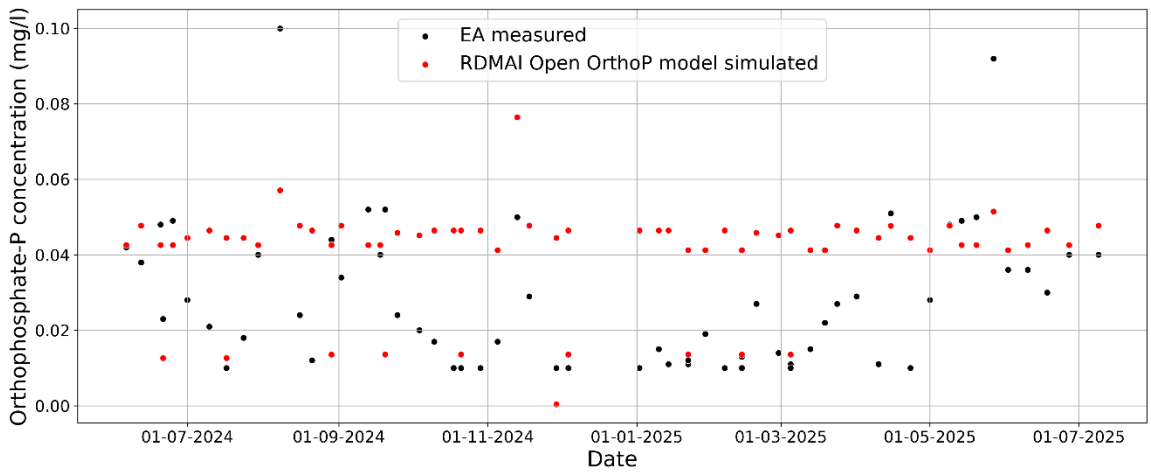

Figure 3 Timeseries plot showing validation output of the RDMAI Open OrthoP Model versus EA measured orthophosphate-P concentration. This example is for the site 'River Piall at Mark's Bridge' in South West England.

5 Case study: Integrated catchment planning to reduce phosphorus concentrations

This hypothetical use case outlines a workflow for a catchment partnership (comprising water companies, NGOs such as The Rivers Trust, and local authorities) to co-design a catchment management plan. The objective is to address a specific regulatory challenge, a river reach failing WFD targets for phosphorus.

By using the RDMAI models in tandem as decision-support tools, stakeholders can move beyond siloed interventions towards collaborative, evidence-based solutions. This workflow does not treat the models as isolated tools; instead, it uses the outputs of one model to refine and validate the inputs of another, creating a progressively more accurate picture of catchment health. If all parties reach consensus on the functioning of the catchment, it builds trust, making it more likely that stakeholders will accept the subsequent evidence on water body health.

5.1.1 Enhancing the hydrological baseline

The workflow begins by improving estimates of pollutant risk calculated by the Open Risk Map. The Open Risk Map is designed to provide an annual average estimation of risk across a catchment; it does not intrinsically represent seasonal variations or specific rainfall events. However, to calculate this annual average, the model requires an estimation of the river's dilution capacity. In its standard configuration, the model uses a simple proxy for this, relying on upstream accumulated river length to estimate flow. The assumption is that river length correlates with discharge volume. While this proxy allows the model to run rapidly without gauged flow data, it introduces significant uncertainty, as it fails to account for the geological differences between catchments, for example, treating a high-volume chalk stream the same as a low-volume clay ditch of the same length.

To rectify this and improve the annual risk estimation, the partnership could deploy the Open Flow Model to replace the flow estimates within the Risk Map. Although the Risk Map does not ingest daily time-series data, the Flow Model could be used to generate flow statistics for the ungauged reaches, such as mean daily flow. Modelled flow would then be substituted into the Risk Map, replacing the rudimentary length-based proxy.

5.1.2 Spatial triage and prioritisation

With the hydrological inputs refined by the Flow Model, the partnership could then run the Open Risk Map to perform a catchment-wide triage. The model functions as a cumulative pollutant routing engine. It ingests high-resolution spatial datasets, specifically CROME 2023 for land cover, Met Office gridded rainfall, and the EA Consents Database for point sources, to build a holistic picture of pollutant loading. By topologically routing these loads down the river network and applying pollutant-

specific decay rates, the model estimates the concentration of Phosphorus at every river node (set to every 200m along the river network).

The model does not just calculate a raw concentration, it converts this into a Risk Value by comparing the predicted annual average concentration against UK TAG Revised Standards for river health. For example, a river node is flagged as 'High Risk' (represented visually as red) if the combined load from upstream sources pushes the concentration beyond an average threshold for 'Poor' status (defined in the model as 0.87 mg/L for Reactive Phosphorus – this can be edited in the code if required, for other river types). This allows the partnership to move beyond guesswork and objectively identify which reaches are more likely to fail regulatory standards.

The partnership can inspect any failing river node to see a breakdown of the pollutant sources, and results can be exported to GIS. For example, the model might reveal that in one reach most risk is driven by sewage discharge from a mapped wastewater treatment works, whereas in a different reach the risk could be dominated by agricultural runoff derived from Farmscoper coefficients applied to specific crop types and livestock. This nuance ensures that the partnership focuses their efforts efficiently, targeting water company asset upgrades in the first reach, while reserving nature-based solutions and farmer engagement for the second reach.

5.1.3 Ground-truthing and validation

With the initial spatial hotspots identified by the Risk Map, the partnership moves to the critical phase of ground-truthing. A theoretical risk map, no matter how detailed, relies on static assumptions about land use and standard discharge consents. To ensure the final catchment management plan is robust, these spatial predictions are validated against measurements taken in the river. However, historical monitoring for phosphorus may be sparse, typically consisting of monthly spot samples that miss pollution spikes. To bridge this gap, the partnership could deploy the Open Orthophosphate Model to generate a higher-frequency, synthetic water quality record that serves as the observed baseline for calibration.

The integration process begins by running the Open Orthophosphate Model at locations where surrogate water quality parameters, such as Dissolved Oxygen, pH, and conductivity, have been measured, but orthophosphate data is missing or infrequent. By identifying complex non-linear relationships between these parameters and nutrient levels, the AI-driven model infills the historical record, creating a time-series of predicted orthophosphate concentrations.

The partnership could compare the Risk Map's estimated annual loading for a specific sub-catchment (a group of nodes) against the loads predicted by the orthophosphate model. If the models align, the Risk Map output is validated.

However, discrepancies trigger a diagnostic step where the temporal resolution of the orthophosphate model adds value to the spatial model. If the Open Orthophosphate Model calculates a high annual load in a sub-catchment that the

Risk Map originally flagged as low risk, the partnership could examine daily patterns in the orthophosphate data to understand why. If the daily data shows the load is independent of rainfall, it suggests a continuous point source; if the load accumulates during winter events, it suggests diffuse runoff. The partnership could use this insight to refine the Risk Map inputs, for example, by adding a missing private sewage treatment works or adjusting runoff coefficients for specific crop types, calibrating the Open Risk map.

5.1.4 'What if' scenario planning

Once the Open Risk Map has been calibrated to match observed nutrient loads, the partnership can transition from diagnosis to strategic solution design. The primary objective is to identify an intervention strategy that brings the river into compliance with WFD targets for phosphorus. This phase moves beyond looking at single sectors in isolation and instead seeks the optimal distribution of effort across all stakeholders; water companies, farmers and urban planners, collaborating with the aim of reducing annual average orthophosphate concentrations to the required target level.

Since the Open Orthophosphate Model relies on historical data and cannot simulate future land-use changes, the calibrated Open Risk Map becomes the primary decision-support tool for scenario planning. The partnership uses the spatial precision of the Risk Map to move away from blanket catchment measures, which are often prohibitively expensive and inefficient. Instead, they can focus on the high-risk areas identified in the triage phase, specifically, the land parcels that the model indicates are generating the highest loading of sediment and particulate phosphorus. For example, the analysis might reveal that 40% of the diffuse risk is generated by just 10% of the catchment area, where steep slopes intersect with arable farming practices near the watercourse.

To test the efficacy of different strategies, the user creates digital interventions by modifying the input data of the Risk Map. This allows them to compare siloed approaches against collaborative ones:

- **Scenario A (Siloed Approach):** The partnership simulates a water company only solution by modifying the Risk Map inputs to simulate upgrading the local wastewater treatment works to tertiary treatment standards. The model runs, potentially revealing that while the point-source load drops, the river still fails the target concentration due to the overwhelming background signal from diffuse agricultural runoff.
- **Scenario B (Collaborative Approach):** The partnership tests a blended strategy. They simulate a minor optimisation at the wastewater treatment works combined with a targeted agricultural intervention, for example, adding riparian buffer zones to high-risk arable farmland.

By comparing the post-intervention risk scores against the baseline, the partnership can quantify the benefit of different interventions. The model might demonstrate that Scenario B achieves the same target concentration as a capital upgrade at the wastewater treatment works but at a fraction of the total cost. This quantifiable output provides the rigorous evidence base required to unlock multi-sector funding, demonstrating to regulators that funding upstream nature-based solutions could provide a defensible and more cost-effective pathway to compliance.

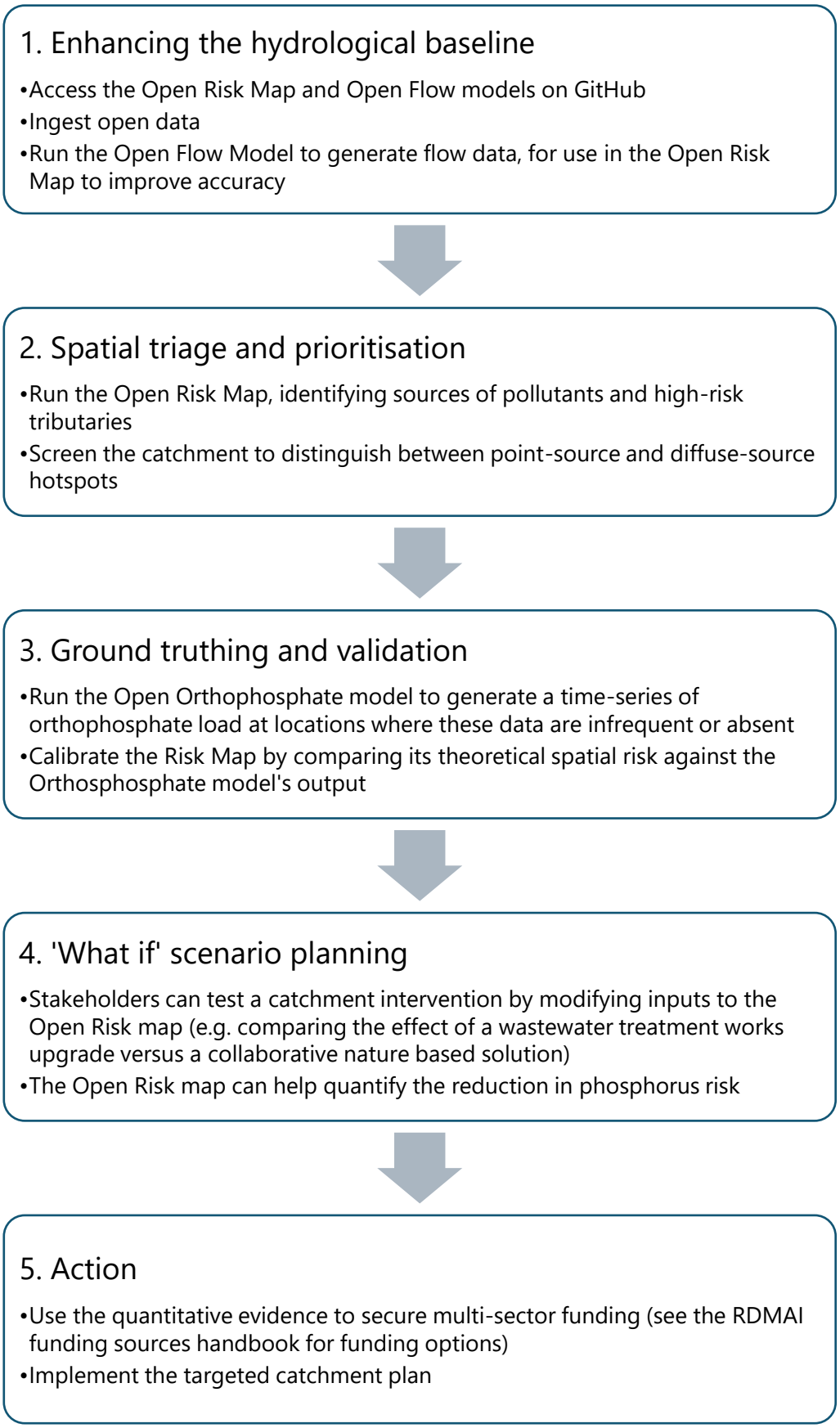

Figure 4 Example workflow for integrated catchment planning to reduce phosphorus pollution using the RDMAI models

6 Case study: Ouse Burn integrated catchment planning

To further explore the potential of the RDMAI models within a real-world setting, discussions were held with catchment stakeholders from the Environment Agency and The Tyne Rivers Trust, focusing on the Ouse Burn catchment in North East England. These discussions highlighted the operational gap between data availability and strategic decision-making, and how the RDMAI models could function as a decision-support tool.

The stakeholders highlighted that despite having access to internal databases and Catchment Data Explorers, prioritising areas for intervention remains largely reactive rather than proactive and data-driven. Currently, the identification of hotspots or deteriorating water bodies often relies on anecdotal evidence, walkovers with parish councils, or local knowledge rather than a systematic synthesis of available data.

While various monitoring efforts exist, including limited Environment Agency spot sampling, water company sondes, and emerging citizen science, there is a lack of capacity to synthesise these disparate sources to determine what all the data means from a water body health perspective. This fragmentation illustrates the silo culture highlighted by the Cunliffe Report (Independent Water Commission, 2025). Because data is often held within organisational boundaries, coordination can become passive, waiting for opportunities or partner interest rather than strategically targeting areas of greatest need. As the Cunliffe Report argues, moving away from this reactive approach requires a shift towards a comprehensive systems planning framework, where data from regulators, utilities, and citizen scientists is actively integrated to drive targeted, evidence-based catchment management.

The stakeholders identified the Open Risk Map as a potential tool for visualising catchment pressures. By providing a map that flags potential risk hotspots, working beyond the boundaries of organisational data silos, the model allows catchment partnerships to coordinate their resources more efficiently.

- **Justifying Investment:** A primary barrier to securing funding (e.g., from the National Lottery or Defra) is the reliance on qualitative evidence. Stakeholders noted that current bids often rely on "probable" or "suspected" causes. The RDMAI models offer the ability to strengthen these bids by moving from anecdotal justification to quantitative prediction.
- **Scenario Planning:** The ability to simulate interventions by adjusting the input data (e.g. "If we invest X amount, the river outcome will be Y") was identified as a potential gamechanger for large-scale restoration projects. However, it must be noted that, at present, adjusting these inputs is not a simple process, as it would require the user to directly edit the underlying model code. Despite this technical barrier, the capability helps partnerships to

perform cost-benefit analyses, demonstrating that integrated, large-scale interventions may deliver better value than piecemeal projects.

- **Stakeholder Engagement:** The models can act as a communication aid when engaging with landowners. Rather than approaching landowners based solely on their willingness to participate, partnerships can prioritise engagement based on modelled risk, using the visual outputs to demonstrate potential improvements.

The discussions revealed a distinct separation between regulatory reporting and catchment coordination. For statutory purposes, the Environment Agency has strict protocols regarding data and model usage. However, for catchment coordination and partnership work, there is significant appetite for using resources like the RDMAI models as a decision support tool to guide investigations, provided they are treated as indicative rather than absolute.

Crucially, the technical accessibility of the models remains the most significant barrier to adoption. Stakeholders emphasised that even within specialist organisations, staff often lack the time, software permissions, or Python skills required to run code-based models. For these tools to be "used in anger" for routine optioneering and funding bids, they must be transitioned into a user-friendly interface (such as a web-based dashboard or GIS viewer) that allows non-technical users to manipulate parameters and view outputs without requiring data scientist support. The RDMAI team has developed proposals to take this forward in a follow-on project.

7 Value Propositions

Deploying the RDMAI models for integrated catchment planning goes beyond regulatory compliance, offering a framework for improved efficiency and stakeholder alignment. By helping replace fragmented data with a unified, open-source evidence base, the models support partners to validate viable potential solutions to water body health challenges. This facilitates a shift from adversarial silos to shared delivery, unlocking cross-sector co-funding opportunities for taking action.

While industry-standard tools like SAGIS (Source Apportionment GIS) remain the benchmark for final statutory reporting, they are often cost-prohibitive and technically rigid for routine optioneering. Unlike proprietary platforms that demand expensive annual seat licenses or specialist consultant operation, the RDMAI suite is open-source and free to deploy, lowering the barrier to entry for iterative scenario planning.

This accessibility allows the RDMAI models to function as a powerful strategic screening tool. Rather than commissioning resource-intensive, full-scale simulations for every sub-catchment, stakeholders can deploy RDMAI for rapid, region-wide assessments. This triage approach identifies priority catchments where complex dynamics exist, allowing water companies and regulators to reserve their high-

fidelity, costly modelling resources for the specific areas where they add the most value.

Whilst it is difficult to calculate a cost saving through using the RDMAI models, the potential savings are significant. The latest Water Industry National Environment Programme (WINEP) sets out a £22.1bn investment in the environment[1]. For 2025/2026, the Environment Agency has a total budget over £189 million for tackling water pollution[2]. A 1% saving on these budgets, through more efficient data access, easier scenario testing and more efficient monitoring, equates to £222M.

8 Limitations

While the use case offers an example of how the RDMAI models could be used for integrated catchment planning, users must be aware of specific constraints within the models.

The Open Risk Map is designed to provide an annual average estimation of risk. Consequently, it does not intrinsically represent seasonal variations, acute low-flow events, or rainfall-driven pollution spikes, except in so far as these factors influence the annual mean. Understanding these seasonal dynamics may be an essential component of certain integrated catchment planning activities. Furthermore, the model relies on open data sources (such as the consents database) which introduces uncertainty, as legal limits may not reflect actual discharge volumes. This uncertainty is likely to be inconsistent across all pollution sources, and therefore could lead to a systematic under-representation of the impact from certain sources relative to others. Similarly, diffuse pollution estimates assume a uniform distribution of livestock, which simplifies the complex spatial heterogeneity of real-world farming. For orthophosphate simulations, the model obtained an average relative root mean squared error of 165%, making it comparable to UKWIR SAGIS simulations (see the benchmarking report for full details). The risk estimations could be improved by substituting the model's rudimentary flow approximation with measured gauge data.

While the RDMAI Open Flow Model serves as a check on hydrological dynamics, it is not universally applicable. Presently, flow can only be simulated at CAMELS-GB catchments, all of which have flow gauges, so a user wishing to simulate flow outside of these catchments would need to amend the model script and undertake additional calculations (e.g., creating the CAMELS-GB data or equivalent for ungauged catchments). Furthermore, the model's ability to extrapolate outside its training data is constrained – additional training is likely to be needed. Validation

[1] https://www.gov.uk/government/news/environment-agency-secures-record-commitments-from-water-sector, accessed 09/02/2026.

[2] https://www.watermagazine.co.uk/2025/07/08/largest-ever-budget-for-tackling-water-pollution-handed-to-the-environment-agency/, accessed 09/02/2026.

indicates that while the model provides reasonable estimates (NSE of 0.54 to 0.62) of average daily flow for approximately 80% of catchments assessed, it under-predicted flow dynamics in the remaining minority. Preliminary analysis by the Rivers Trust suggests that the Open Flow Model performs better in some river morphologies than others – therefore accuracy is likely to vary depending on the river being investigated. The specific hydrological or geological reasons for these shortcomings are currently undetermined, meaning the model cannot yet be universally applied to every ungauged catchment without a degree of verification. Model improvements would require advanced Python and GIS skills.

Stakeholders must interpret the Open Orthophosphate model's outputs with the understanding that it predicts soluble reactive orthophosphate, not Total Phosphorus (TP). This may limit some applications of the model for integrated catchment planning. Additionally, the training data included only samples with a maximum concentration of 0.5mg/L. This means the model cannot extrapolate concentrations for highly polluted events exceeding this threshold; outputs are effectively capped at 0.5mg/L. Additionally, because the model is trained on low-frequency EA spot samples (typically 1-2 per month), it likely under-represents acute pollution spikes driven by storm events. Model validation against unseen EA water quality data (June 2024 to July 2025) obtained a Normalised Root Mean Square Error (NRMSE) of 10.15%. Using high-frequency EA sonde data as model input and validation against citizen science measures of orthophosphate-P also indicated that the RDMAI Open OrthoP model struggles to represent high concentrations of orthophosphate-P. During the benchmarking exercise, the Relative Root Mean Square Error (RRMSE) was calculated as a comparison of the mean model estimates with the mean EA measures. RRMSE ranged between 2.11% and 6.43% for different regions of England (see the benchmarking report for full details).

Each model has an accompanying report, discussing how the model was trained, evaluated and the validation approach used. The model reports can be viewed here.

9 Next steps and opportunities

Currently, the use case workflow for integrated catchment planning requires a human-in-the-loop to manually bridge the gap between spatial outputs (from the Risk Map) and numerical outputs (for the Flow and Orthophosphate models). A valuable development step would be the creation of an automated wrapper or unified dashboard that links these tools. This would allow for seamless data pipelining, for example, hydrological statistics generated by the Open Flow Model could be automatically substituted into the Open Risk Map to refine dilution estimates, or paired with Open Orthophosphate Model concentration outputs to automate the calculation of pollutant loads. This integration would reduce the technical burden on users and minimise the risk of manual data handling errors.

Looking further ahead, there is scope to expand the geographic and functional versatility of the suite. While the models are currently trained on English datasets (EA water quality data and CAMELS-GB), testing and retraining them on data from the devolved administrations would allow for a consistent approach to river management across the UK. However, this expansion is contingent upon the availability of comparable, high-resolution open datasets from regulatory bodies in Scotland, Wales, and Northern Ireland.

Given the open-source principle used to develop the models, it is anticipated that the community of users will drive innovation that continues to evolve the models into increasingly effective tools.

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